FEU Institute of Technology
COLLEGE OF ENGINEERING • COLLEGE OF COMPUTER STUDIES

Computer System and Architecture

Machine Problem

MIDTERM

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Name of Student	Name of Professor
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Data Performed	Date Submitted

PROGRAM DESCRIPTION:
This program is a menu-driven Certificate of Registration (COR) system developed using NASM 32-bit Assembly Language. It allows users to input and display student information, calculate tuition fees, save the complete COR details into a file, read saved records, and exit the program.

COMPILATION:

```
@zjvalenzuela-zzzzz ➜/workspaces/FLAB3/LABEXER3 (main) $ nasm -f elf32 labexer3.asm -o labexer3.o
@zjvalenzuela-zzzzz ➜/workspaces/FLAB3/LABEXER3 (main) $ ld -m elf_i386 -e _start labexer3.o -o labexer3
@zjvalenzuela-zzzzz ➜/workspaces/FLAB3/LABEXER3 (main) $ ./labexer3
```

PROCESS/INPUT OUTPUT (CHOICE 1)

```
COR -> Input, Output, Procedure, Macro
=======================================
                COR
=======================================
1. Input  -> Input the details (1st Part, 3rd Part) TF
2. Output -> Display the full info
3. Write  - Write the full details into a file
4. Read   - Reads the content of the file
5. Exit

Enter your choice:
1

--- ENTER STUDENT INFORMATION ---
Enrollment Status (e.g., ENROLLED):
ENROLLED
Student Number:
202510947
Full Name:
Zack Gabriel J. Valenzuela
Phone Number:
09667218351
ZIP Code:
1420
College:
COMPUTER STUDIES
Program:
BSCSSE
Year Level:
1
Term (e.g., 1st term):
 1st term
School Year (e.g., SY 2026-2027):
SY 26-27
Complete Address:
BLK 42-A BELGRADE STREET, VISTA VERDE NORTH LLANO ROAD

--- SUBJECTS AND UNITS ---
Enter 0 units if you are not taking a subject.
1. CCS0021  - INFORMATION MANAGEMENT (LEC)
   Units for this subject (whole number):
2
2. CCS0021L - INFORMATION MANAGEMENT (LAB)
   Units for this subject (whole number):
```

```
3. CS0001   - DISCRETE STRUCTURES I
   Units for this subject (whole number):
3
4. CS0003   - COMPUTER SYSTEMS & ARCHITECTURE
   Units for this subject (whole number):
2
5. CS0003L  - COMPUTER SYSTEMS & ARCHITECTURE (LAB)
   Units for this subject (whole number):
1
6. CS0070   - OBJECT ORIENTED PROGRAMMING
   Units for this subject (whole number):
2
7. CS0070L  - OBJECT ORIENTED PROGRAMMING-LAB
   Units for this subject (whole number):
1
8. GED0021  - SPECIALIZED ENGLISH PROGRAM 2
   Units for this subject (whole number):
3
9. GED0055  - MATHEMATICAL ANALYSIS
   Units for this subject (whole number):
3
10. GED0081  - COLLEGE PHYSICS 1 LECTURE
   Units for this subject (whole number):
2
11. GED0081L - COLLEGE PHYSICS 1 LABORATORY
   Units for this subject (whole number):
1

Student information and subject units recorded!

COR -> Input, Output, Procedure, Macro
=======================================
                COR
=======================================
1. Input  -> Input the details (1st Part, 3rd Part) TF
2. Output -> Display the full info
3. Write  - Write the full details into a file
4. Read   - Reads the content of the file
5. Exit

Enter your choice:
```

PROCESS/INPUT OUTPUT (CHOICE 2)

```
=======================================
                COR
=======================================
1. Input  -> Input the details (1st Part, 3rd Part) TF
2. Output -> Display the full info
3. Write  - Write the full details into a file
4. Read   - Reads the content of the file
5. Exit

Enter your choice:
2

=========================================
      SOLAR / COR TUITION CALCULATOR
=========================================


--- STUDENT INFORMATION SUMMARY ---
Enrollment Status: ENROLLED
Student Number: 202510947
Name: Zack Gabriel J. Valenzuela
Phone: 09667218351
ZIP Code: 1420
College: COMPUTER STUDIES
Program: BSCSSE
Year Level: 1
Term:  1st term
School Year: SY 26-27
Address: BLK 42-A BELGRADE STREET, VISTA VERDE NORTH LLANO ROAD

--- SUBJECT UNIT SUMMARY ---
1. CCS0021  - INFORMATION MANAGEMENT (LEC)  Units: 2
2. CCS0021L - INFORMATION MANAGEMENT (LAB)  Units: 1
3. CS0001   - DISCRETE STRUCTURES I  Units: 3
4. CS0003   - COMPUTER SYSTEMS & ARCHITECTURE  Units: 2
5. CS0003L  - COMPUTER SYSTEMS & ARCHITECTURE (LAB)  Units: 1
6. CS0070   - OBJECT ORIENTED PROGRAMMING  Units: 2
7. CS0070L  - OBJECT ORIENTED PROGRAMMING-LAB  Units: 1
8. GED0021  - SPECIALIZED ENGLISH PROGRAM 2  Units: 3
9. GED0055  - MATHEMATICAL ANALYSIS  Units: 3
10. GED0081  - COLLEGE PHYSICS 1 LECTURE  Units: 2
11. GED0081L - COLLEGE PHYSICS 1 LABORATORY  Units: 1
```

```
--- TUITION FEE SUMMARY ---
Total units: 21
Unit price: PHP 1000
Tuition (total units x PHP 1,000): PHP 21000
Laboratory subjects taken: 4
Laboratory fee (PHP 2,500 each): PHP 10000
TOTAL ESTIMATED TUITION + LAB FEES: PHP 31000

Note: This estimate excludes miscellaneous, ID validation,
and any other fees not specified in the calculation.

COR -> Input, Output, Procedure, Macro
=======================================
                COR
=======================================
1. Input  -> Input the details (1st Part, 3rd Part) TF
2. Output -> Display the full info
3. Write  - Write the full details into a file
4. Read   - Reads the content of the file
5. Exit

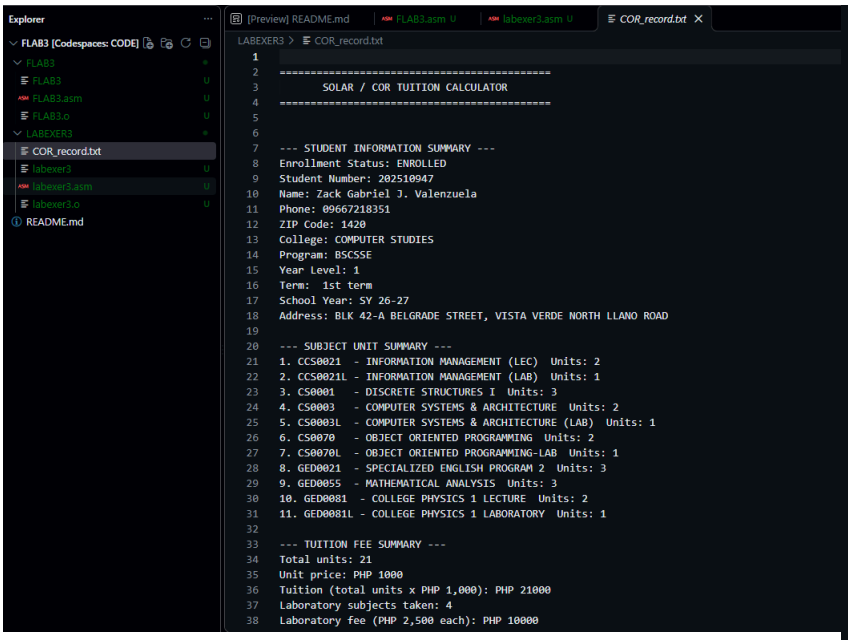
Enter your choice:
```

PROCESS/INPUT OUTPUT (CHOICE 3)

```
COR -> Input, Output, Procedure, Macro
========================================
                COR
========================================
1. Input  -> Input the details (1st Part, 3rd Part) TF
2. Output -> Display the full info
3. Write  - Write the full details into a file
4. Read   - Reads the content of the file
5. Exit

Enter your choice:
3
```

Txt file




```
7  --- STUDENT INFORMATION SUMMARY ---
8  Enrollment Status: ENROLLED
9  Student Number: 202510947
10 Name: Zack Gabriel J. Valenzuela
11 Phone: 09667218351
12 ZIP Code: 1420
13 College: COMPUTER STUDIES
14 Program: BSCSSE
15 Year Level: 1
16 Term:  1st term
17 School Year: SY 26-27
18 Address: BLK 42-A BELGRADE STREET, VISTA VERDE NORTH LLANO ROAD
19
20 --- SUBJECT UNIT SUMMARY ---
21 1. CCS0021  - INFORMATION MANAGEMENT (LEC)  Units: 2
22 2. CCS0021L - INFORMATION MANAGEMENT (LAB)  Units: 1
23 3. CS0001   - DISCRETE STRUCTURES I  Units: 3
24 4. CS0003   - COMPUTER SYSTEMS & ARCHITECTURE  Units: 2
25 5. CS0003L  - COMPUTER SYSTEMS & ARCHITECTURE (LAB)  Units: 1
26 6. CS0070   - OBJECT ORIENTED PROGRAMMING  Units: 2
27 7. CS0070L  - OBJECT ORIENTED PROGRAMMING-LAB  Units: 1
28 8. GED0021  - SPECIALIZED ENGLISH PROGRAM 2  Units: 3
29 9. GED0055  - MATHEMATICAL ANALYSIS  Units: 3
30 10. GED0081  - COLLEGE PHYSICS 1 LECTURE  Units: 2
31 11. GED0081L - COLLEGE PHYSICS 1 LABORATORY  Units: 1
32
33 --- TUITION FEE SUMMARY ---
34 Total units: 21
35 Unit price: PHP 1000
36 Tuition (total units x PHP 1,000): PHP 21000
37 Laboratory subjects taken: 4
38 Laboratory fee (PHP 2,500 each): PHP 10000
39 TOTAL ESTIMATED TUITION + LAB FEES: PHP 31000
40
41 Note: This estimate excludes miscellaneous, ID validation,
42 and any other fees not specified in the calculation.
43
```

PROCESS/INPUT OUTPUT (CHOICE 4)

```
Enter your choice:
4

--- READING SAVED COR FILE ---

========================================
       SOLAR / COR TUITION CALCULATOR
========================================


--- STUDENT INFORMATION SUMMARY ---
Enrollment Status: ENROLLED
Student Number: 202510947
Name: Zack Gabriel J. Valenzuela
Phone: 09667218351
ZIP Code: 1420
College: COMPUTER STUDIES
Program: BSCSSE
Year Level: 1
Term:  1st term
School Year: SY 26-27
Address: BLK 42-A BELGRADE STREET, VISTA VERDE NORTH LLANO ROAD

--- SUBJECT UNIT SUMMARY ---
1. CCS0021  - INFORMATION MANAGEMENT (LEC)  Units: 2
2. CCS0021L - INFORMATION MANAGEMENT (LAB)  Units: 1
3. CS0001   - DISCRETE STRUCTURES I  Units: 3
4. CS0003   - COMPUTER SYSTEMS & ARCHITECTURE  Units: 2
5. CS0003L  - COMPUTER SYSTEMS & ARCHITECTURE (LAB)  Units: 1
6. CS0070   - OBJECT ORIENTED PROGRAMMING  Units: 2
7. CS0070L  - OBJECT ORIENTED PROGRAMMING-LAB  Units: 1
8. GED0021  - SPECIALIZED ENGLISH PROGRAM 2  Units: 3
9. GED0055  - MATHEMATICAL ANALYSIS  Units: 3
10. GED0081  - COLLEGE PHYSICS 1 LECTURE  Units: 2
11. GED0081L - COLLEGE PHYSICS 1 LABORATORY  Units: 1

--- TUITION FEE SUMMARY ---
Total units: 21
Unit price: PHP 1000
Tuition (total units x PHP 1,000): PHP 21000
Laboratory subjects taken: 4
Laboratory fee (PHP 2,500 each): PHP 10000
TOTAL ESTIMATED TUITION + LAB FEES: PHP 31000

Note: This estimate excludes miscellaneous, ID validation,
and any other fees not specified in the calculation.

COR -> Input, Output, Procedure, Macro
========================================
                  COR
========================================
1. Input  -> Input the details (1st Part, 3rd Part) TF
2. Output -> Display the full info
3. Write  - Write the full details into a file
4. Read   - Reads the content of the file
5. Exit

Enter your choice:
```

PROCESS/INPUT OUTPUT (CHOICE 5)

```
Enter your choice:
5

Thank you for using the COR program!
@zjvalenzuela-zzzzz →/workspaces/FLAB3/LABEXER3 (main) $
```

SOURCE CODE:

```
; ZACK GABRIEL J. VALENZUELA
; Solar menu cor
; NASM 32-bit Assembly for Linux

GLOBAL _start

%macro PRINT 2
    pushad
    mov eax, 4
    mov ebx, [out_fd]
    mov ecx, %1
    mov edx, %2
    int 0x80
    popad
%endmacro

%macro ASK_STRING 4
    PRINT %1, %2
    push dword %4
    push dword %3
    call read_line
    add esp, 8
%endmacro

%macro SHOW_FIELD 3
    PRINT %1, %2
    push dword %3
    call print_string
    add esp, 4
    PRINT newline, newline_len
%endmacro

SECTION .data

    heading db 10, '============================================', 10
            db '       SOLAR / COR TUITION CALCULATOR', 10
            db '============================================', 10, 10
    heading_len equ $ - heading

    student_heading db 10, '--- ENTER STUDENT INFORMATION ---', 10
    student_heading_len equ $ - student_heading
```

```
p_status db 'Enrollment Status (e.g., ENROLLED):', 10
p_status_len equ $ - p_status
p_student_no db 'Student Number:', 10
p_student_no_len equ $ - p_student_no
p_name db 'Full Name:', 10
p_name_len equ $ - p_name
p_phone db 'Phone Number:', 10
p_phone_len equ $ - p_phone
p_zip db 'ZIP Code:', 10
p_zip_len equ $ - p_zip
p_college db 'College:', 10
p_college_len equ $ - p_college
p_program db 'Program:', 10
p_program_len equ $ - p_program
p_year_level db 'Year Level:', 10
p_year_level_len equ $ - p_year_level
p_term db 'Term (e.g., 1st term):', 10
p_term_len equ $ - p_term
p_school_year db 'School Year (e.g., SY 2026-2027):', 10
p_school_year_len equ $ - p_school_year
p_address db 'Complete Address:', 10
p_address_len equ $ - p_address

summary_heading db 10, '--- STUDENT INFORMATION SUMMARY ---', 10
summary_heading_len equ $ - summary_heading

l_status db 'Enrollment Status: '
l_status_len equ $ - l_status
l_student_no db 'Student Number: '
l_student_no_len equ $ - l_student_no
l_name db 'Name: '
l_name_len equ $ - l_name
l_phone db 'Phone: '
l_phone_len equ $ - l_phone
l_zip db 'ZIP Code: '
l_zip_len equ $ - l_zip
l_college db 'College: '
l_college_len equ $ - l_college
l_program db 'Program: '
l_program_len equ $ - l_program
l_year_level db 'Year Level: '
l_year_level_len equ $ - l_year_level
l_term db 'Term: '
```

```
l_term_len equ $ - l_term
l_school_year db 'School Year: '
l_school_year_len equ $ - l_school_year
l_address db 'Address: '
l_address_len equ $ - l_address

subjects_heading db 10, '--- SUBJECTS AND UNITS ---', 10
                 db 'Enter 0 units if you are not taking a subject.', 10
subjects_heading_len equ $ - subjects_heading

units_prompt db '   Units for this subject (whole number): ', 10
units_prompt_len equ $ - units_prompt

unit_summary_heading db 10, '--- SUBJECT UNIT SUMMARY ---', 10
unit_summary_heading_len equ $ - unit_summary_heading

units_label db '  Units: '
units_label_len equ $ - units_label

fee_heading db 10, '--- TUITION FEE SUMMARY ---', 10
fee_heading_len equ $ - fee_heading

label_total_units db 'Total units: '
label_total_units_len equ $ - label_total_units

label_unit_price db 'Unit price: PHP '
label_unit_price_len equ $ - label_unit_price

label_tuition db 'Tuition (total units x PHP 1,000): PHP '
label_tuition_len equ $ - label_tuition

label_lab_subjects db 'Laboratory subjects taken: '
label_lab_subjects_len equ $ - label_lab_subjects

label_lab_fee db 'Laboratory fee (PHP 2,500 each): PHP '
label_lab_fee_len equ $ - label_lab_fee

label_total_due db 'TOTAL ESTIMATED TUITION + LAB FEES: PHP '
label_total_due_len equ $ - label_total_due

footer db 10, 'Note: This estimate excludes miscellaneous, ID validation,', 10
       db 'and any other fees not specified in the calculation.', 10
footer_len equ $ - footer
```

```
newline db 10
newline_len equ $ - newline

zero_text db '0', 0

subj1  db '1. CCS0021  - INFORMATION MANAGEMENT (LEC)', 0
subj2  db '2. CCS0021L - INFORMATION MANAGEMENT (LAB)', 0
subj3  db '3. CS0001   - DISCRETE STRUCTURES I', 0
subj4  db '4. CS0003   - COMPUTER SYSTEMS & ARCHITECTURE', 0
subj5  db '5. CS0003L  - COMPUTER SYSTEMS & ARCHITECTURE (LAB)', 0
subj6  db '6. CS0070   - OBJECT ORIENTED PROGRAMMING', 0
subj7  db '7. CS0070L  - OBJECT ORIENTED PROGRAMMING-LAB', 0
subj8  db '8. GED0021  - SPECIALIZED ENGLISH PROGRAM 2', 0
subj9  db '9. GED0055  - MATHEMATICAL ANALYSIS', 0
subj10 db '10. GED0081  - COLLEGE PHYSICS 1 LECTURE', 0
subj11 db '11. GED0081L - COLLEGE PHYSICS 1 LABORATORY', 0

subject_table dd subj1, subj2, subj3, subj4, subj5, subj6
              dd subj7, subj8, subj9, subj10, subj11

lab_flags db 0, 1, 0, 0, 1, 0, 1, 0, 0, 0, 1

php_unit_price equ 1000
php_lab_price equ 2500

menu db 10, 'COR -> Input, Output, Procedure, Macro', 10
     db '=========================================', 10
     db '               COR', 10
     db '=========================================', 10
     db '1. Input  -> Input the details (1st Part, 3rd Part) TF', 10
     db '2. Output -> Display the full info', 10
     db '3. Write  - Write the full details into a file', 10
     db '4. Read   - Reads the content of the file', 10
     db '5. Exit', 10, 10
     db 'Enter your choice: ', 10
menu_len equ $ - menu

invalid_choice db 10, 'Invalid choice. Please enter 1 to 5.', 10
invalid_choice_len equ $ - invalid_choice

invalid_units db 'Invalid units. Enter a whole number from 0 to 30:', 10
invalid_units_len equ $ - invalid_units
```

```
    input_done db 10, 'Student information and subject units recorded!', 10
    input_done_len equ $ - input_done

    no_information db 10, 'Please select Option 1 to enter COR information first.', 10
    no_information_len equ $ - no_information

    saved_notice db 10, 'COR details saved to COR_record.txt', 10
    saved_notice_len equ $ - saved_notice

    save_failed db 10, 'Error: unable to write COR_record.txt', 10
    save_failed_len equ $ - save_failed

    read_heading db 10, '--- READING SAVED COR FILE ---', 10
    read_heading_len equ $ - read_heading

    read_failed db 10, 'Error: COR_record.txt cannot be opened. Save it with Option 3
first.', 10
    read_failed_len equ $ - read_failed

    exit_message db 10, 'Thank you for using the COR program!', 10
    exit_message_len equ $ - exit_message

    file_name db 'COR_record.txt', 0

SECTION .bss

    status_buf      resb 128
    student_no_buf  resb 128
    name_buf        resb 128
    phone_buf       resb 128
    zip_buf         resb 128
    college_buf     resb 128
    program_buf     resb 128
    year_level_buf  resb 128
    term_buf        resb 128
    school_year_buf resb 128
    address_buf     resb 128

    integer_buffer resb 32
    units_array resd 11

    total_units resd 1
```

```
    lab_subject_count resd 1
    tuition_amount resd 1
    lab_amount resd 1
    total_amount resd 1

    digit_buffer resb 16
    out_fd resd 1
    saved_fd resd 1
    has_data resb 1
    char_buf resb 1
    file_buffer resb 4096

SECTION .text

_start:
    mov dword [out_fd], 1
    mov byte [has_data], 0

.menu_loop:
    PRINT menu, menu_len
    call read_integer
    jnc .valid_choice
    cmp eax, -1
    je .exit
    jmp .invalid

.valid_choice:
    cmp eax, 1
    je .input
    cmp eax, 2
    je .output
    cmp eax, 3
    je .write
    cmp eax, 4
    je .read
    cmp eax, 5
    je .exit

.invalid:
    PRINT invalid_choice, invalid_choice_len
    jmp .menu_loop

.input:
```

```
    call input_details
    mov byte [has_data], 1
    PRINT input_done, input_done_len
    jmp .menu_loop

.output:
    cmp byte [has_data], 1
    jne .need_input
    call display_report
    jmp .menu_loop

.write:
    cmp byte [has_data], 1
    jne .need_input
    call save_report
    jmp .menu_loop

.read:
    call read_saved_file
    jmp .menu_loop

.need_input:
    PRINT no_information, no_information_len
    jmp .menu_loop

.exit:
    jmp quit_program

quit_program:
    PRINT exit_message, exit_message_len
    mov eax, 1
    xor ebx, ebx
    int 0x80

input_details:
    PRINT student_heading, student_heading_len

    ASK_STRING p_status, p_status_len, status_buf, 128
    ASK_STRING p_student_no, p_student_no_len, student_no_buf, 128
    ASK_STRING p_name, p_name_len, name_buf, 128
    ASK_STRING p_phone, p_phone_len, phone_buf, 128
    ASK_STRING p_zip, p_zip_len, zip_buf, 128
    ASK_STRING p_college, p_college_len, college_buf, 128
```

```
    ASK_STRING p_program, p_program_len, program_buf, 128
    ASK_STRING p_year_level, p_year_level_len, year_level_buf, 128
    ASK_STRING p_term, p_term_len, term_buf, 128
    ASK_STRING p_school_year, p_school_year_len, school_year_buf, 128
    ASK_STRING p_address, p_address_len, address_buf, 128

    mov dword [total_units], 0
    mov dword [lab_subject_count], 0

    PRINT subjects_heading, subjects_heading_len

    xor esi, esi

.subject_loop:
    cmp esi, 11
    jge .subjects_done

    mov eax, [subject_table+esi*4]
    push eax
    call print_string
    add esp, 4

    PRINT newline, newline_len
    PRINT units_prompt, units_prompt_len

.get_units:
    call read_integer
    jnc .check_units
    cmp eax, -1
    je quit_program
    jmp .invalid_units

.check_units:
    cmp eax, 30
    ja .invalid_units

    mov [units_array+esi*4], eax
    add [total_units], eax

    cmp eax, 0
    je .next_subject

    cmp byte [lab_flags+esi], 1
```

```
    jne .next_subject

    inc dword [lab_subject_count]

.next_subject:
    inc esi
    jmp .subject_loop

.invalid_units:
    PRINT invalid_units, invalid_units_len
    jmp .get_units

.subjects_done:
    mov eax, [total_units]
    imul eax, php_unit_price
    mov [tuition_amount], eax

    mov eax, [lab_subject_count]
    imul eax, php_lab_price
    mov [lab_amount], eax

    mov eax, [tuition_amount]
    add eax, [lab_amount]
    mov [total_amount], eax
    ret

display_report:
    PRINT heading, heading_len
    PRINT summary_heading, summary_heading_len

    SHOW_FIELD l_status, l_status_len, status_buf
    SHOW_FIELD l_student_no, l_student_no_len, student_no_buf
    SHOW_FIELD l_name, l_name_len, name_buf
    SHOW_FIELD l_phone, l_phone_len, phone_buf
    SHOW_FIELD l_zip, l_zip_len, zip_buf
    SHOW_FIELD l_college, l_college_len, college_buf
    SHOW_FIELD l_program, l_program_len, program_buf
    SHOW_FIELD l_year_level, l_year_level_len, year_level_buf
    SHOW_FIELD l_term, l_term_len, term_buf
    SHOW_FIELD l_school_year, l_school_year_len, school_year_buf
    SHOW_FIELD l_address, l_address_len, address_buf

    PRINT unit_summary_heading, unit_summary_heading_len
```

```
    xor esi, esi

.subject_loop:
    cmp esi, 11
    jge .fees

    mov eax, [subject_table+esi*4]
    push eax
    call print_string
    add esp, 4

    PRINT units_label, units_label_len

    push dword [units_array+esi*4]
    call print_uint
    add esp, 4

    PRINT newline, newline_len

    inc esi
    jmp .subject_loop

.fees:
    PRINT fee_heading, fee_heading_len

    PRINT label_total_units, label_total_units_len
    push dword [total_units]
    call print_uint
    add esp, 4
    PRINT newline, newline_len

    PRINT label_unit_price, label_unit_price_len
    push dword php_unit_price
    call print_uint
    add esp, 4
    PRINT newline, newline_len

    PRINT label_tuition, label_tuition_len
    push dword [tuition_amount]
    call print_uint
    add esp, 4
    PRINT newline, newline_len
```

```
    PRINT label_lab_subjects, label_lab_subjects_len
    push dword [lab_subject_count]
    call print_uint
    add esp, 4
    PRINT newline, newline_len

    PRINT label_lab_fee, label_lab_fee_len
    push dword [lab_amount]
    call print_uint
    add esp, 4
    PRINT newline, newline_len

    PRINT label_total_due, label_total_due_len
    push dword [total_amount]
    call print_uint
    add esp, 4
    PRINT newline, newline_len

    PRINT footer, footer_len
    ret

save_report:
    mov eax, 8
    mov ebx, file_name
    mov ecx, 420
    int 0x80

    test eax, eax
    js .fail

    mov [saved_fd], eax
    mov [out_fd], eax

    call display_report

    mov dword [out_fd], 1

    mov eax, 6
    mov ebx, [saved_fd]
    int 0x80

    PRINT saved_notice, saved_notice_len
```

```
    ret

.fail:
    PRINT save_failed, save_failed_len
    ret

read_saved_file:
    mov eax, 5
    mov ebx, file_name
    xor ecx, ecx
    xor edx, edx
    int 0x80

    test eax, eax
    js .fail

    mov [saved_fd], eax
    PRINT read_heading, read_heading_len

.read_loop:
    mov eax, 3
    mov ebx, [saved_fd]
    mov ecx, file_buffer
    mov edx, 4096
    int 0x80

    test eax, eax
    jle .close

    mov edi, eax
    xor esi, esi

.write_loop:
    mov eax, 4
    mov ebx, 1
    lea ecx, [file_buffer+esi]
    mov edx, edi
    sub edx, esi
    int 0x80

    test eax, eax
    jle .close
```

```
    add esi, eax
    cmp esi, edi
    jb .write_loop
    jmp .read_loop

.close:
    mov eax, 6
    mov ebx, [saved_fd]
    int 0x80
    ret

.fail:
    PRINT read_failed, read_failed_len
    ret

print_string:
    push ebp
    mov ebp, esp
    push ebx
    push esi

    mov esi, [ebp+8]
    xor edx, edx

.count_chars:
    cmp byte [esi+edx], 0
    je .write_chars
    inc edx
    jmp .count_chars

.write_chars:
    test edx, edx
    jz .print_done

    mov eax, 4
    mov ebx, [out_fd]
    mov ecx, esi
    int 0x80

.print_done:
    pop esi
    pop ebx
    mov esp, ebp
```

```
    pop ebp
    ret

read_line:
    push ebp
    mov ebp, esp
    push ebx
    push esi
    push edi

    mov edi, [ebp+8]
    mov esi, [ebp+12]
    dec esi
    xor ebx, ebx

.read_next:
    push ebx
    mov eax, 3
    xor ebx, ebx
    mov ecx, char_buf
    mov edx, 1
    int 0x80
    pop ebx

    cmp eax, 0
    jg .got_char

    cmp ebx, 0
    jne .finished

    mov byte [edi], 0
    mov eax, -1
    jmp .done

.got_char:
    mov al, [char_buf]
    cmp al, 10
    je .finished
    cmp al, 13
    je .read_next
    cmp ebx, esi
    jae .read_next
```

```
    mov [edi+ebx], al
    inc ebx
    jmp .read_next

.finished:
    mov byte [edi+ebx], 0
    mov eax, ebx

.done:
    pop edi
    pop esi
    pop ebx
    mov esp, ebp
    pop ebp
    ret

read_integer:
    push ebp
    mov ebp, esp
    push ebx
    push esi
    push edx

    push dword 32
    push dword integer_buffer
    call read_line
    add esp, 8

    cmp eax, -1
    je .eof

    cmp eax, 0
    jle .invalid

    xor eax, eax
    mov esi, integer_buffer

.parse_digit:
    mov bl, [esi]
    test bl, bl
    jz .valid

    cmp bl, '0'
```

```
    jb .invalid

    cmp bl, '9'
    ja .invalid

    cmp eax, 99999
    ja .invalid

    imul eax, eax, 10
    sub bl, '0'
    movzx edx, bl
    add eax, edx

    inc esi
    jmp .parse_digit

.valid:
    clc
    jmp .done

.eof:
    mov eax, -1
    stc
    jmp .done

.invalid:
    xor eax, eax
    stc

.done:
    pop edx
    pop esi
    pop ebx
    mov esp, ebp
    pop ebp
    ret

print_uint:
    push ebp
    mov ebp, esp
    push ebx
    push esi
    push edi
```

```
    mov eax, [ebp+8]

    test eax, eax
    jnz .convert_number

    push dword zero_text
    call print_string
    add esp, 4
    jmp .number_done

.convert_number:
    mov edi, digit_buffer
    add edi, 15
    mov byte [edi], 0
    mov ebx, 10

.convert_loop:
    xor edx, edx
    div ebx

    add dl, '0'
    dec edi
    mov [edi], dl

    test eax, eax
    jnz .convert_loop

    push edi
    call print_string
    add esp, 4

.number_done:
    pop edi
    pop esi
    pop ebx
    mov esp, ebp
    pop ebp
    ret
```